

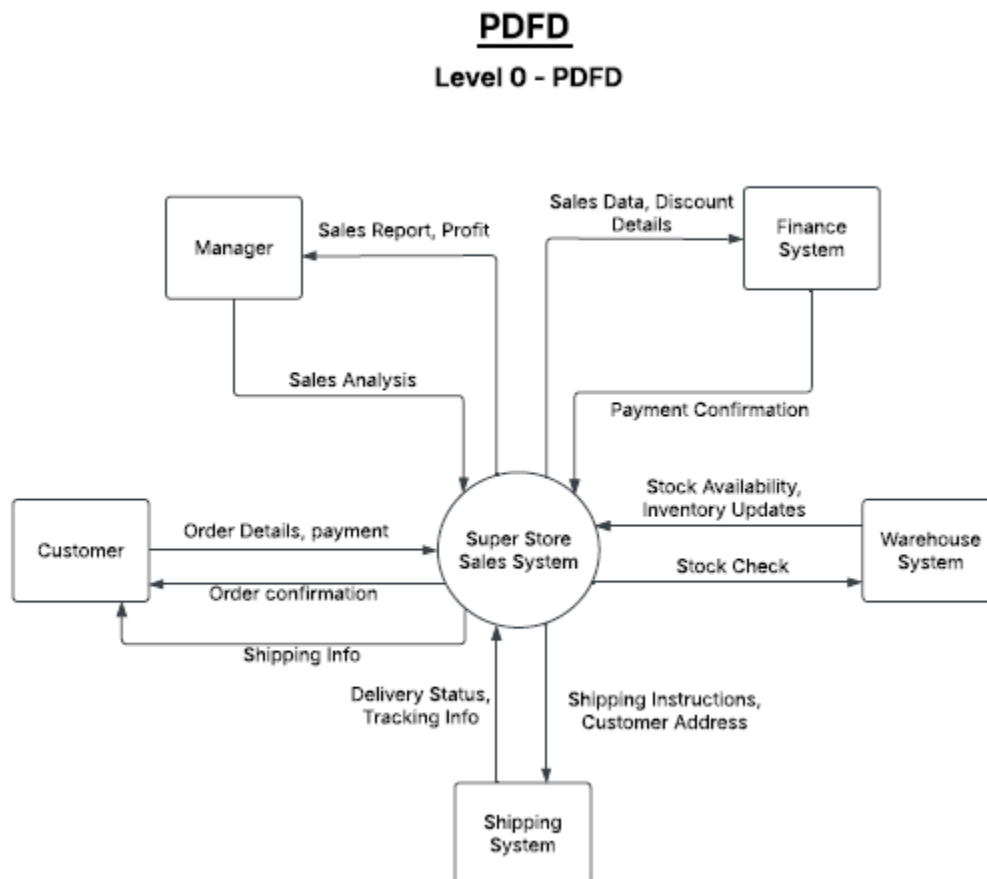
Applied A.I. Solutions Foundations of Data Management

Lab Exercise 2

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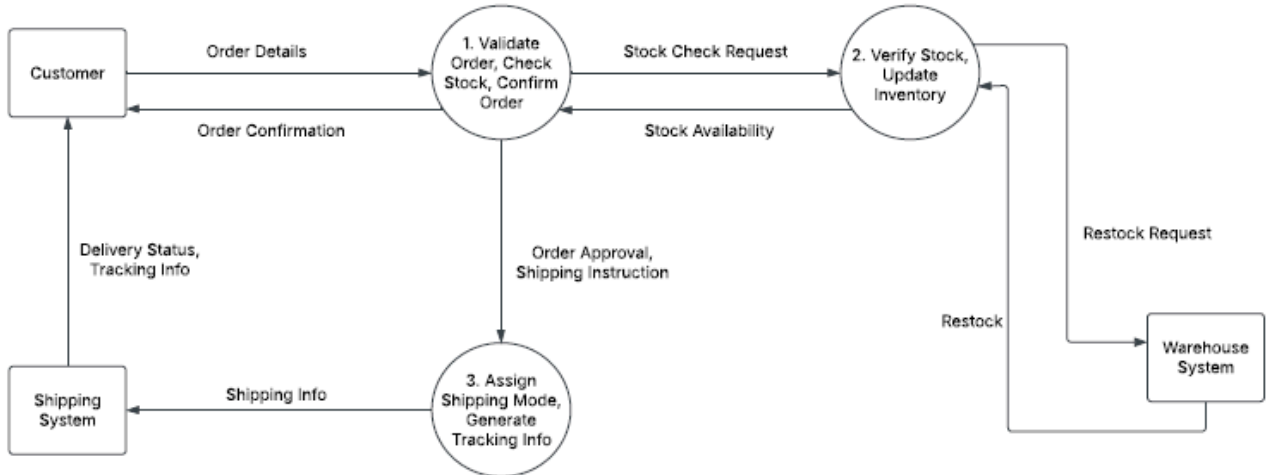
Data Flow Diagram

Level 0



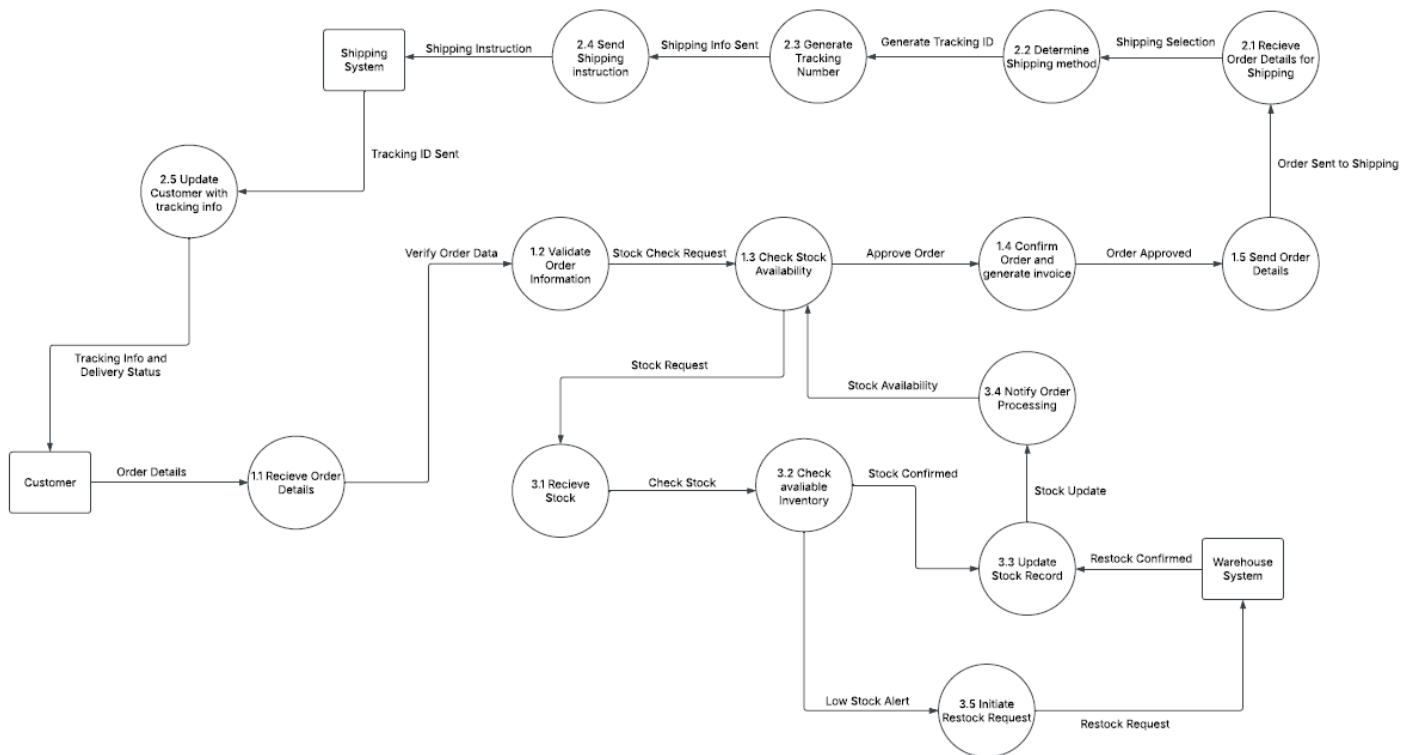
Level 1

Level 1 - PDFD (just order)

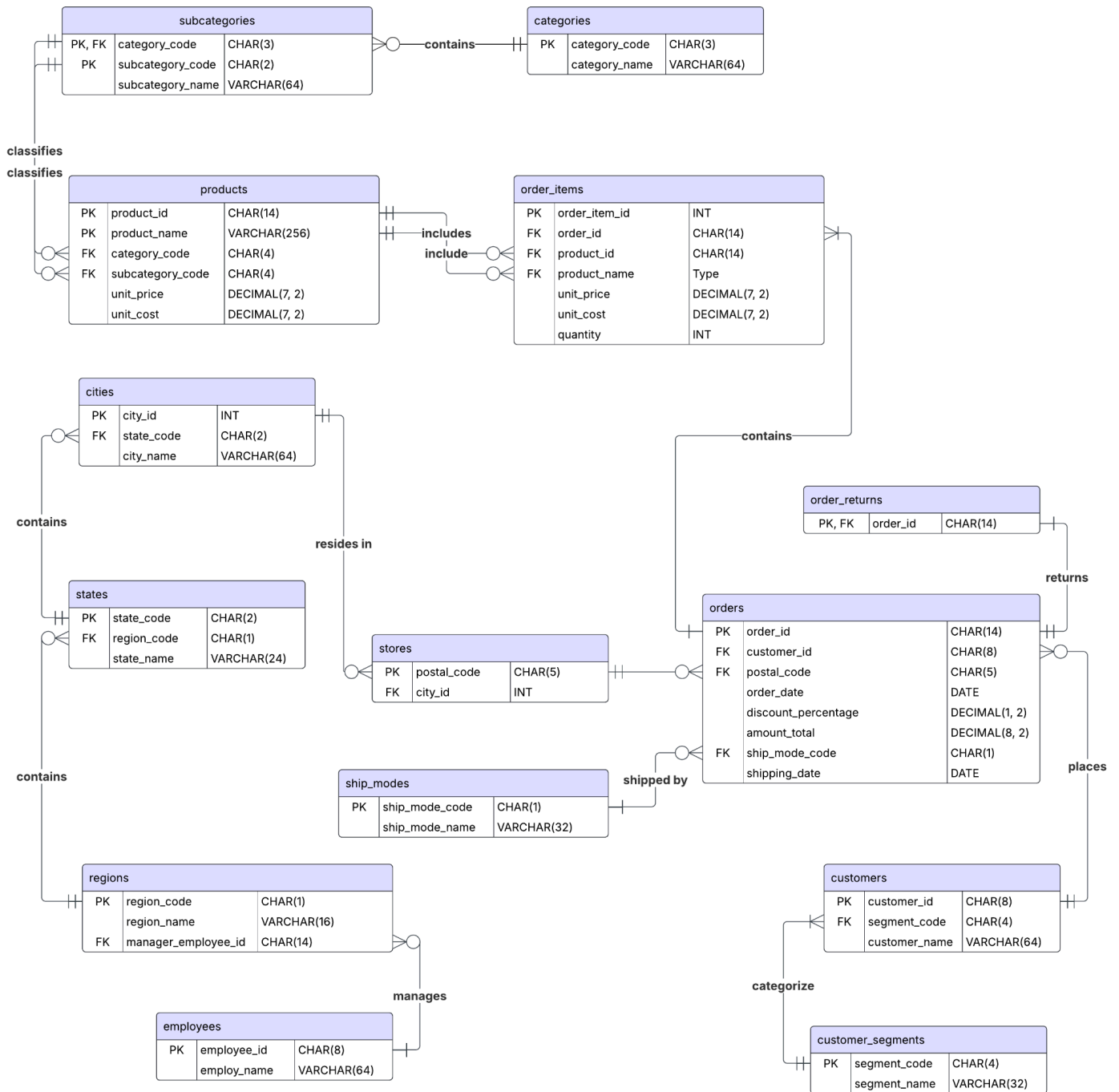


Level 2

Level 2 - PDFD (just order)



Entity-Relationship Diagram



Assumptions for Normalization and ERD

1. Products and Categories (Categories, Subcategories)

Assumptions

1. Each product must belong to one and only one subcategory.
2. Each subcategory must belong to one and only one main category.
3. A product cannot belong to multiple subcategories at the same time.
4. Each product must have a unique `product_id`—no duplicate product IDs are allowed.
5. The `unit_price` and `unit_cost` of a product cannot be negative.

Impact

- The `products` table must enforce `category_code` and `subcategory_code` as foreign keys (FK) to ensure data consistency.
 - A `CHECK` constraint may be needed for `unit_price` and `unit_cost` to ensure non-negative values.
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2. Orders and Order Items

Assumptions

1. Each order must contain at least one product but can include multiple products.
2. Each order item must be linked to an existing order.
3. The `quantity` field in order items must be at least 1 (no zero or negative values).
4. The `amount_total` field in orders should be calculated automatically and not manually entered.
5. The `discount_percentage` cannot be less and equal to 0% or more than 100%.
6. All orders must be associated with a `customer_id`; anonymous orders are not allowed.
7. The `shipping_date` cannot be earlier than the `order_date`.
8. Returned orders in `order_returns` must already exist in `orders`; nonexistent orders cannot be returned.

Impact

- The `order_items.quantity` field should have a `CHECK(quantity > 0)` constraint.
 - The `orders.discount_percentage` field should have a `CHECK(0 <= discount_percentage <= 100)` constraint.
 - The `orders.shipping_date` field should have a `CHECK(shipping_date >= order_date)` constraint.
 - The `order_returns` table should have a mandatory foreign key reference to `orders`.
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3. Customers and Customer Segments

Assumptions

1. Each customer must belong to a customer segment; no customers should exist without a segment.
2. Each `customer_id` must be unique (no duplicate customers).
3. Customer segments are predefined and cannot be freely chosen by customers.
4. Customer segments have a limited number of categories (e.g., "Individual Customer," "Business Customer," "VIP Customer").
5. A customer segment cannot be deleted if it still has associated customers.

Impact

- The `customers.segment_code` field must be a foreign key referencing `customer_segments`.
 - The `customer_segments.segment_code` field should be unique and serve as a primary key (PK).
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4. Stores, States, and Regions

Assumptions

1. Each store must belong to a specific state (`state_code` cannot be NULL).
2. Each state must belong to a region (`region_code` cannot be NULL).
3. Each region has a designated manager (`manager_employee_id`).
4. Each store has a unique `postal_code`; no two stores share the same postal code.
5. State names must be unique and cannot be NULL.

Impact

- The `stores.state_code` field must be a foreign key (FK) referencing `states`.
 - The `states.region_code` field must be a foreign key (FK) referencing `regions`.
 - The `regions.manager_employee_id` field must be a foreign key (FK) referencing `employees`.
-

5. Employees

Assumptions

1. Each employee must have a unique `employee_id`; duplicate IDs are not allowed.
2. Employee names (`employ_name`) cannot be NULL and must not exceed 64 characters.
3. Managers (`manager_employee_id`) are also employees, meaning `manager_employee_id` must exist in the `employees` table.
4. Each region can have only one manager; multiple managers for a single region are not allowed.

Impact

- The `employees.employee_id` field must be a unique primary key (PK).
 - The `regions.manager_employee_id` field should be a foreign key (FK) referencing `employees.employee_id`.
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6. Shipping Modes

Assumptions

1. Each order must have a shipping mode (`ship_mode_code` cannot be NULL).
2. Shipping modes are predefined and cannot be entered arbitrarily.
3. Multiple shipping modes should be supported, such as “Standard Shipping” and “Express Shipping.”
4. The `ship_mode_code` must be unique and should consist of a single character (CHAR(1)).

Impact

- The `orders.ship_mode_code` field should be a foreign key (FK) referencing `ship_modes`.
- The `ship_modes.ship_mode_code` field should have a unique constraint (PK).